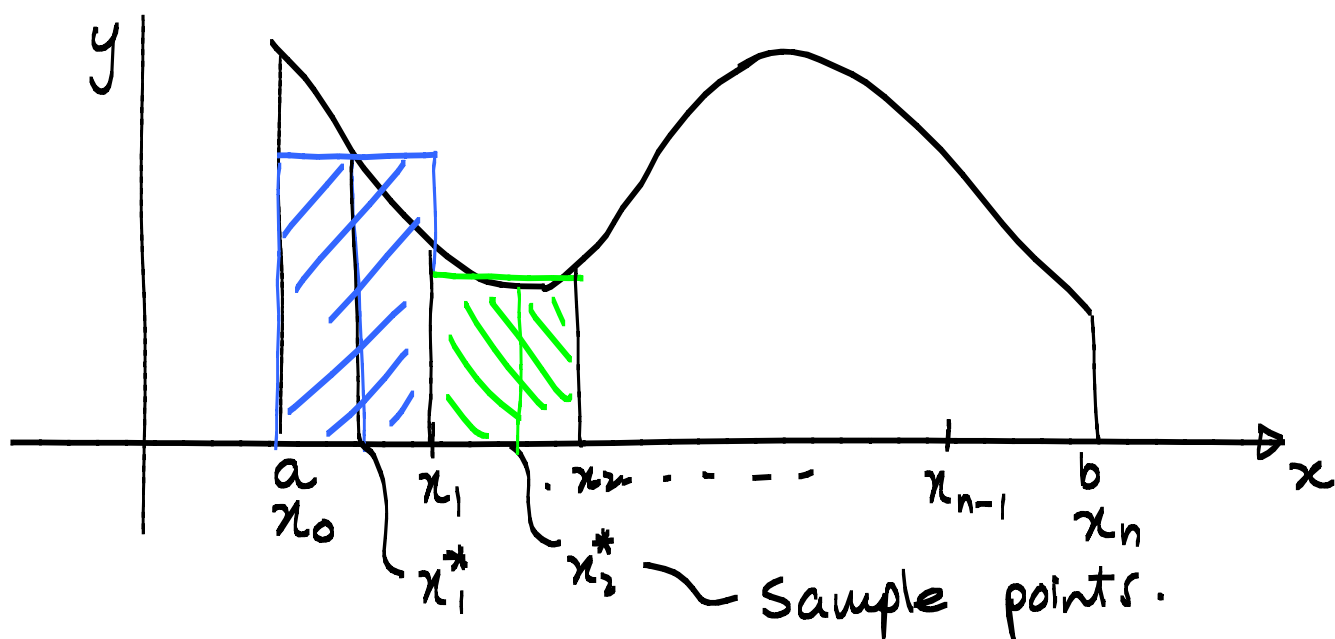


Unit 5.3 The definite integral p 371

Note Title

2014/04/20



$$\Delta x = \frac{b-a}{n}$$

$$A = f(x_1^*)(x_1 - x_0) + f(x_2^*)(x_2 - x_1) + \dots + f(x_n^*)(x_n - x_{n-1})$$

$$A = \sum_{i=1}^n \underline{\underline{f(x_i^*)}} \underline{\underline{\Delta x}}$$

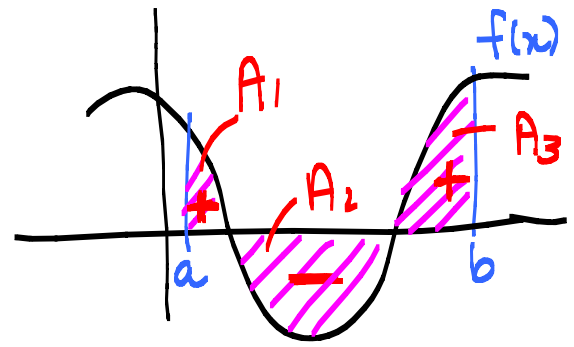
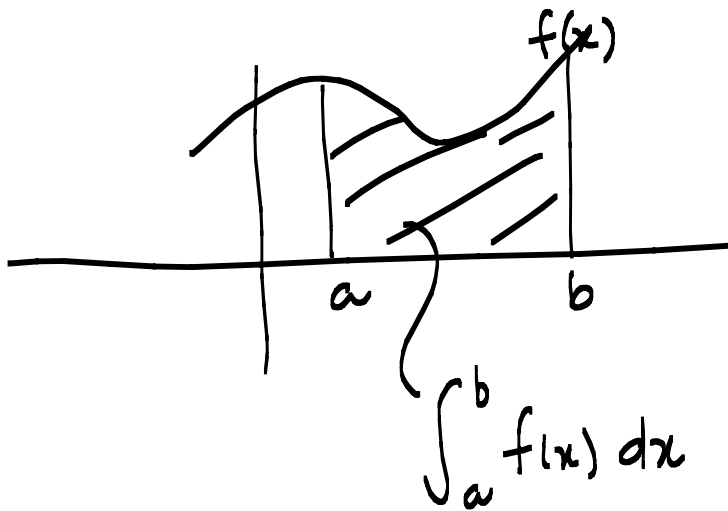
$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \underbrace{\sum_{i=1}^n f(x_i^*) \Delta x}_{\text{Riemann sum}}$$

limits of integration

Integrand

Definite integral

So: If $f(x) \geq 0$ then we have the area under the curve!

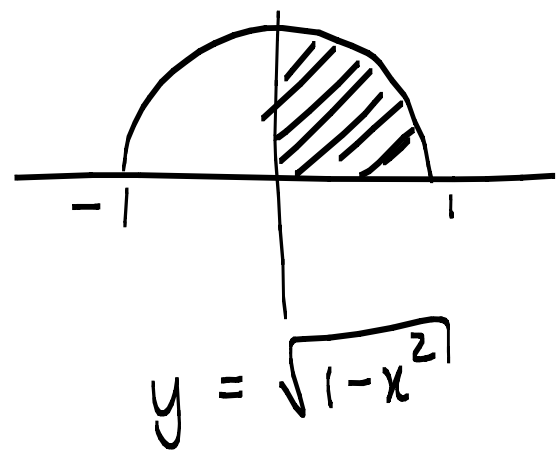


$$\text{Area} = A_1 - A_2 + A_3$$

Determine the integral by using the area of f between the curve and the x-axis.

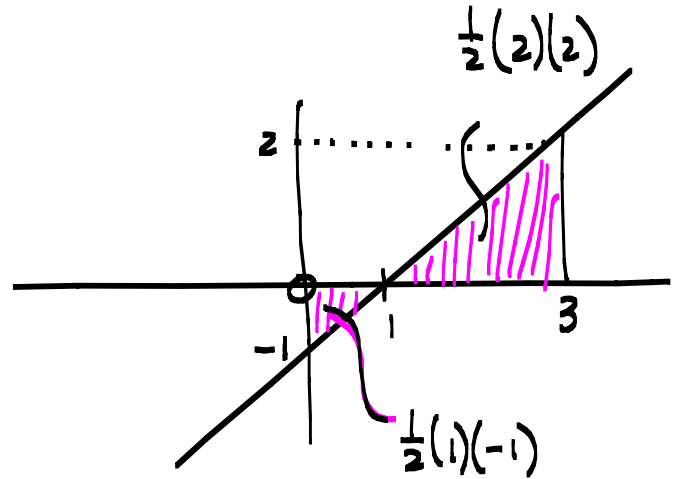
$$1. \int_0^1 \sqrt{1-x^2} dx$$

$$= \frac{1}{4} \pi r^2 = \frac{\pi}{4}$$



$$2. \int_0^3 (x-1) dx$$

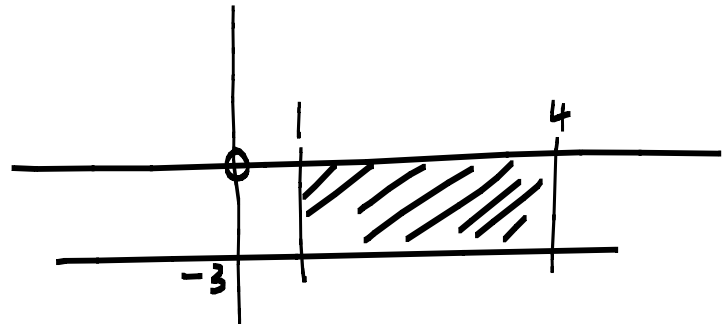
$$= -\frac{1}{2} + 2$$



$$3. \int_1^4 (-3) dx$$

$$= 3(-3)$$

$$= -9$$



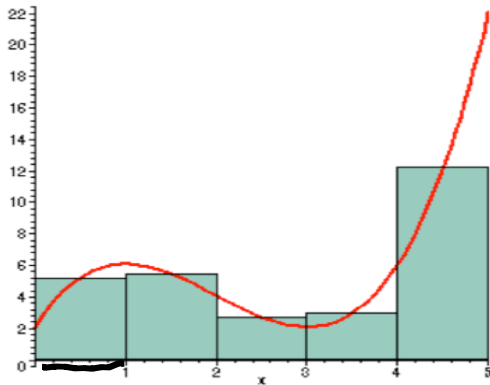
The integral can be negative
whereas the area cannot.

$$\text{Area} = 9$$

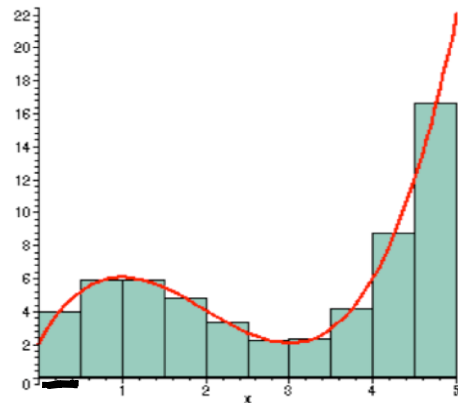
$$\int_1^4 (-3) dx = -9$$

Example:

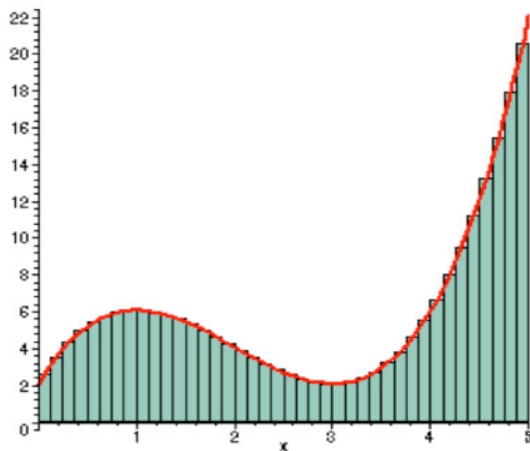
Find the area under the curve of
 $f(x) = x^3 - 6x^2 + 9x + 2$ between
 $x = 0$ and $x = 5$.



$$(f(1/2) + f(3/2) + f(5/2) + f(7/2) + f(9/2)) = 28.125$$



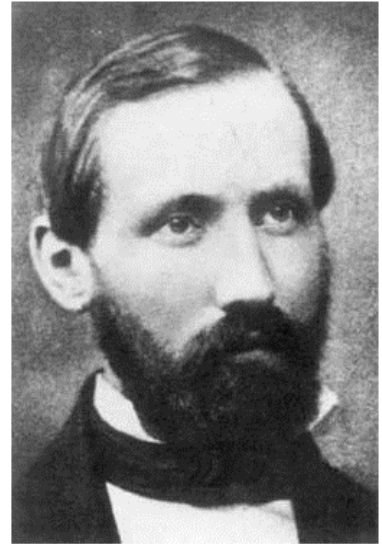
$$(f(1/4) + f(3/4) + f(5/4) + \dots + f(19/4)) = 28.58375$$



$$(f(1/16) + f(3/16) + f(5/16) + \dots + f(39/16)) = 28.74023$$

$$\begin{aligned} & \int_0^5 (x^3 - 6x^2 + 9x + 2) \, dx \\ &= \left(\frac{1}{4} x^4 - 2x^3 + \frac{9}{2} x^2 + 2x \right) \Big|_0^5 \\ &= 28.75 \end{aligned}$$

The method was named after German mathematician [Bernhard Riemann](#). Riemann was born in Germany in September 1826



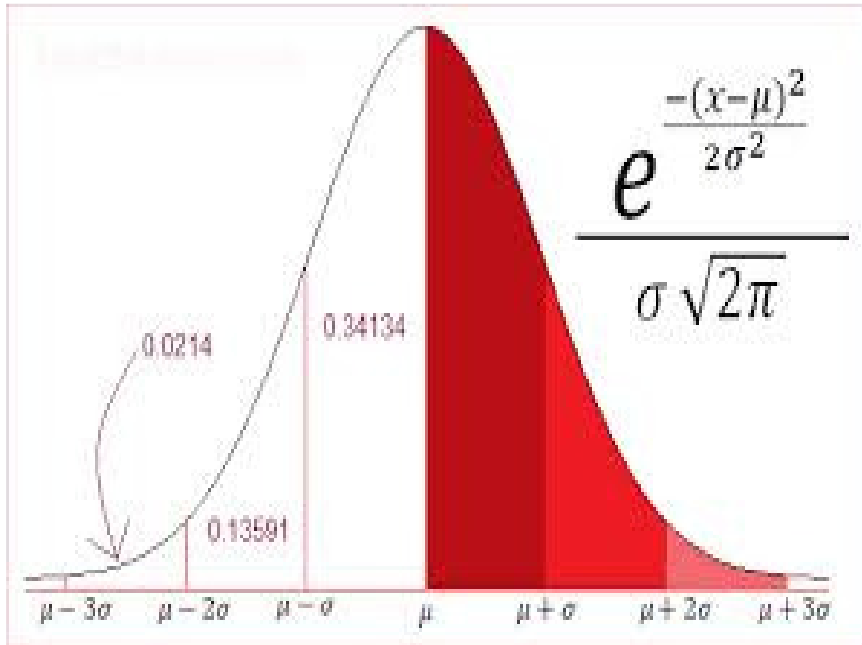
Most famous contribution:
Riemann sum.

Calculus developed in
1660's and 1670's by
Newton and Leibniz.

Integration simply a process reverse
to differentiation — 19th century.

Real world problems.

Gaussian function



$$y = \frac{1}{\sqrt{2\pi}} e^{-(x-\mu)^2/2\sigma}$$

μ = Mean

σ = Standard Deviation

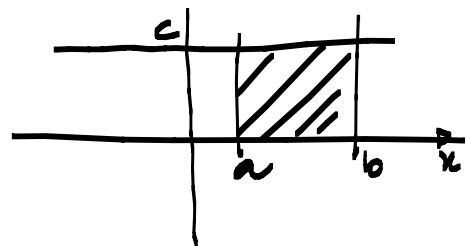
$\pi \approx 3.14159$

$e \approx 2.71828$

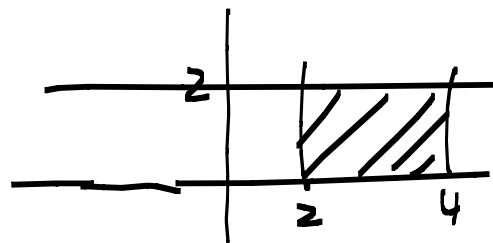
Gaussian functions are widely used in [statistics](#) where they describe the [normal distributions](#), in [signal processing](#) where they serve to define [Gaussian filters](#), in [image processing](#) where two-dimensional Gaussians are used for [Gaussian blurs](#), and in [mathematics](#) where they are used to solve [heat equations](#) and [diffusion equations](#) and to define the [Weierstrass transform](#).

Properties of the integral p 373

$$1. \int_a^b c \, dx = c(b-a)$$



$$\int_2^4 2 \, dx = \underline{4}$$



$$\int_2^4 2 \, dx = 2x \Big|_2^4$$

$$= 2(4) - 2(2) \quad **$$

Prove!!!

$$= 8 - 4 = \underline{4}$$

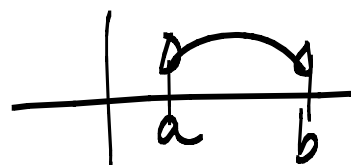
$$\int_2^4 2 \, dx = (2x + c) \Big|_2^4$$

$$= 2(4) + \cancel{c} - (2(2) + \cancel{c})$$

$$2. \int_a^b (f(x) \pm g(x)) \, dx = \int_a^b f(x) \, dx \pm \int_a^b g(x) \, dx$$

$$3. \int_a^b c f(x) \, dx = c \int_a^b f(x) \, dx$$

$$4. \int_a^b f(x) \, dx = - \int_b^a f(x) \, dx$$



$$\Delta x = \frac{b-a}{n}$$

$$5. \int_a^b f(x) dx = 0$$

Examples:

$$\begin{aligned} 1) \int_{-1}^2 e^{5x} dx &= \frac{1}{5} e^{5x} \Big|_{-1}^2 \\ &= \frac{1}{5} e^{5(2)} - \frac{1}{5} e^{5(-1)} \\ &= \frac{1}{5} \left(e^{10} - \frac{1}{e^5} \right) \end{aligned}$$

$$\begin{aligned} 2) \int_{-2}^1 x^2 e^{x^3} dx &= \frac{1}{3} e^{x^3} \Big|_{-2}^1 \\ &= \frac{1}{3} (e^1 - e^{-8}) \end{aligned}$$

$$\begin{aligned} 3) \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin x dx &= -\cos x \Big|_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \\ &= -\cos\left(\frac{\pi}{2}\right) + \cos\left(-\frac{\pi}{2}\right) \\ &= 0 \end{aligned}$$

Find the area of the function $f(x) = \sin x$ between the curve and the x -axis between $x = -\frac{\pi}{2}$ and $x = \frac{\pi}{2}$

$$A = A_1 + A_2$$

Finish it !!

